

THO CHU, Viet Nam

WMO: 489160

Lat: 9.299N

Lon: 103.503E

Elev: 24

StdP: 101.04

Time Zone: 7.00 (E07)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.8	24.3	18.5	13.4	25.5	19.6	14.4	25.9	10.9	27.0	9.8	26.7	2.5	270	0.358

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	5.5	33.6	27.9	32.8	27.6	32.1	27.3	28.5	32.3	28.1	31.8	27.7	31.1	1.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.3	23.2	30.9	27.0	22.8	30.5	26.7	22.4	30.0	92.0	32.6	90.2	32.0	88.6	31.4	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.3	6.0	DB	22.5	35.5	1.3	1.1	21.5	36.4	20.7	37.0	20.0	37.7	19.1	38.5	(4)
(5)			WB	19.6	29.3	1.6	0.4	18.5	29.6	17.5	29.8	16.6	30.1	15.5	30.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.1	27.1	27.6	28.8	29.8	29.5	28.5	27.9	27.9	27.6	27.6	27.7	27.2	(6)	
	DBStd	1.43	0.89	0.92	0.89	1.19	1.57	1.42	1.30	1.31	1.30	1.18	0.90	0.99	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	6605	530	493	582	593	603	554	554	556	527	546	532	534	(10)	
	CDD18.3	3563	272	259	323	343	345	304	296	298	277	288	282	276	(11)	
	CDH23.3	37270	2426	2377	3407	4056	4143	3417	3121	3100	2766	2893	2914	2651	(12)	
	CDH26.7	10539	373	496	1077	1690	1712	1132	830	809	613	661	673	474	(13)	
(14) Wind	WSAvg	2.0	2.6	2.5	2.3	1.6	1.4	1.6	1.7	1.6	1.7	1.8	2.7	2.7	(14)	
(15) Precipitation	PrecAvg	2281	36	20	42	86	184	274	322	266	364	337	236	112	(15)	
	PrecMax	2941	117	87	325	314	317	575	446	390	659	730	402	302	(16)	
	PrecMin	1761	0	0	0	0	5	72	111	126	190	122	91	18	(17)	
	PrecStd	295	33	28	71	78	81	124	94	76	124	153	90	75	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.7	31.6	33.1	34.8	35.1	33.4	32.4	32.5	32.2	32.0	31.1	30.8	(19)	
		MCWB	26.2	26.1	27.0	28.1	28.4	28.4	28.0	28.1	28.0	27.6	27.0	26.8	(20)	
	2%	DB	29.8	30.7	32.1	33.6	33.7	32.2	31.3	31.4	31.1	30.9	30.4	30.0	(21)	
		MCWB	25.5	25.7	26.5	27.6	28.1	27.9	27.5	27.6	27.4	27.1	26.7	26.2	(22)	
	5%	DB	29.1	30.0	31.4	32.7	32.7	31.4	30.6	30.6	30.3	30.1	29.8	29.3	(23)	
		MCWB	25.2	25.4	26.2	27.3	27.8	27.5	27.1	27.2	27.0	26.7	26.4	25.6	(24)	
	10%	DB	28.5	29.2	30.6	31.9	31.7	30.5	29.8	29.8	29.5	29.4	29.2	28.7	(25)	
		MCWB	24.8	25.0	25.9	27.0	27.5	27.1	26.8	26.8	26.6	26.4	26.0	25.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.7	26.7	27.5	28.5	29.0	28.8	28.5	28.5	28.2	27.9	27.6	27.2	(27)	
		MCDB	30.0	30.6	32.1	33.5	33.6	32.4	31.6	31.7	31.4	31.2	30.5	30.2	(28)	
	2%	WB	26.0	26.1	26.9	27.9	28.4	28.2	27.8	27.8	27.6	27.4	27.0	26.6	(29)	
		MCDB	29.1	30.0	31.2	32.6	32.7	31.5	30.6	30.8	30.6	30.3	29.9	29.3	(30)	
	5%	WB	25.6	25.6	26.5	27.5	28.0	27.7	27.3	27.4	27.1	27.0	26.6	26.1	(31)	
		MCDB	28.4	29.2	30.5	31.9	32.0	30.7	29.9	30.1	29.9	29.7	29.2	28.7	(32)	
	10%	WB	25.2	25.2	26.1	27.1	27.6	27.2	26.9	26.9	26.7	26.6	26.2	25.7	(33)	
		MCDB	27.8	28.4	29.8	31.1	31.1	29.8	29.3	29.4	29.0	28.9	28.5	28.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	3.6	4.5	5.2	5.5	4.8	4.1	3.9	4.0	3.8	3.8	3.5	3.2	(35)	
		MCDBR	4.3	5.1	5.8	6.3	6.1	5.3	5.1	5.2	5.1	4.8	4.2	4.1	(36)	
	5% WB	MCWBR	2.0	2.2	2.2	2.2	2.3	2.4	2.5	2.6	2.4	2.4	2.1	2.0	(37)	
		MCWBR	4.1	5.0	5.8	6.1	5.7	5.0	4.8	5.1	4.9	4.7	4.1	4.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.508	0.511	0.519	0.480	0.439	0.445	0.447	0.444	0.436	0.459	0.471	0.486	(40)	
		taud	2.159	2.134	2.166	2.299	2.459	2.463	2.441	2.439	2.450	2.383	2.324	2.245	(41)	
	Ebn at Noon		806	818	816	838	856	841	842	857	874	854	835	817	(42)	
		Edh at Noon	152	160	157	136	113	111	114	117	116	123	128	137	(43)	
(44) All-Sky Solar Radiation	RadAvg		5.20	5.91	6.26	6.30	5.13	4.56	4.44	4.63	4.48	4.62	4.99	4.94	(44)	
	RadStd		0.29	0.31	0.33	0.40	0.44	0.38	0.49	0.34	0.50	0.46	0.30	0.27	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		+0.30	+0.45	N/A	N/A	+0.18	+0.18	N/A	N/A	N/A	N/A

Nomenclature: See separate page